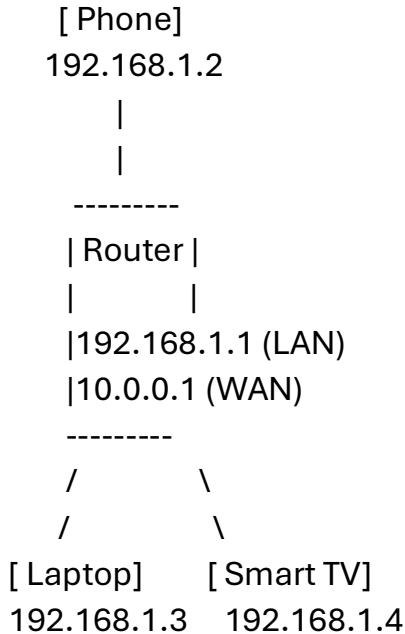


IP ROUTING AND ROUTING PROTOCOLS

Network Routing and OSPF Configuration

1. Simple Network Diagram (with IP Addresses)



All devices are in the same LAN: 192.168.1.0/24

2. Packet Travel (Laptop → Friend's Computer in Another Subnet)

Example

- Your laptop: **192.168.1.3**
- Friend PC: **192.168.2.5**
- Router:
 - LAN: **192.168.1.1**
 - WAN: **192.168.2.1**

Step-by-Step Process

1. Laptop checks IP → **192.168.2.5** (different subnet)
2. Sends packet to the **default gateway (192.168.1.1)**

3. Router receives the packet

4. Router checks the routing table:

192.168.1.0/24 → LAN interface

192.168.2.0/24 → WAN interface

5. Router forwards the packet to the **192.168.2.0 network**

6. Friend's computer receives the packet

Final Path

Laptop → Router → Friend PC

3. Routing Table (Zomato-Style Delivery Network)

Destination Network	Subnet Mask	Next Hop	Interface
10.1.0.0	255.255.255.0	Direct	Delhi
10.2.0.0	255.255.255.0	10.1.0.2	Delhi
10.3.0.0	255.255.255.0	10.1.0.3	Mumbai

Meaning

- Delhi, Mumbai, and Bangalore represent different delivery hubs.
- The router decides the best path, similar to assigning a delivery driver to reach a destination efficiently.

4. Static vs Dynamic Routing

Static Routing

Pros

- Simple to set up
- No extra network traffic
- More secure

Cons

- Manual updates needed

- Not scalable (bad for big apps like Zomato)
- Fails if a link break

Dynamic Routing (RIP/OSPF)

Pros

- Automatically updates routes
- Good for large systems (like Uber/Zomato)
- Handles failures automatically

Cons

- More complex
- Uses CPU and memory
- Can cause overhead traffic

5. Simple OSPF Configuration and Explanation

Sample Configuration

```
router ospf 1
```

```
network 192.168.1.0 0.0.0.255 area 0
```

```
network 192.168.2.0 0.0.0.255 area 0
```

Explanation

- **router ospf 1** → Starts OSPF process number 1
- **network 192.168.1.0 0.0.0.255 area 0** → Includes the first network in OSPF
- **0.0.0.255** → Wildcard mask (represents a /24 subnet)
- **area 0** → Backbone area of OSPF (main routing area)
- **network 192.168.2.0 0.0.0.255 area 0** → Adds another network to OSPF